25-26 IY5609/IY4609/IY3609: Digital Forensics

Main content blocks

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- 19 January 2026, 8:30 AM

Olga Angelopoulou

Welcome to Digital Forensics (IY3609/ IY4609/ IY5609)

Older topics ...

Topic outline

- Collapse all

Digital Forensics (IY5609 / IY4609 / IY3609)

Welcome to the Digital Forensics module page for the 2025-26 academic year.

Module Leader	Dr Olga Angelopoulou
Office	Bedford 1-05
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Office Hours	Wednesday 1-3pm - By appointment on MS Team
Lecture Location	BOURNE-LT1
Lecture Time	Wednesday 10am-1pm

Module Summary:

The module is offered as a postgraduate optional module in the MSc in Information Security, and is also offered at the undergraduate level (IY3609 and IY4609) with adaptations in the assessment. The actual content is not differentiated between the module codes.

Module Learning Outcomes:

LO1 - Discuss legal requirements for acquiring, analysing, and presenting digital evidence in a (UK) court of law.

LO2 - Gather host-based evidence from file systems (esp. Microsoft windows-based), email, and audit records.
LO3 - Discuss methods and requirements for obtaining network-based evidence.
LO4 - Apply analysis techniques for malware infiltration, exfiltration, and obfuscation mechanisms.

Indicative Lecture Weekly Plan:

	Theme	Topic
Week 1	Introduction	1. Introduction to th 2. Digital forensics
Week 2		Principles and Prac 1. Legislation and R 2. Investigation me
Week 3	File Systems	Data storage basics
Week 4		File systems & files
Week 5	Analysis	Host forensics and
Week 6		Host forensics and
Week 7	Specialist themes	Digital forensics in
Week 8		Network forensics
Week 9		Malware Analysis f
Week 10		Mobile forensic pri
Week 11		Forensic Process, D

Reading List

Assessment:

1. Moodle Quiz - 20%

The quiz is a timed (1h) assessment in which (typically) four problem solving questions need to be answered concisely.
The quiz is scheduled to take place on 18 February.

2. Exam - 80%

Invigilated exam during the Summer assessment period.

Block Mode Week Schedule - Block Mode Students ONLY

- **SANS SIFT Workstation**

This module uses the SANS SIFT Workstation for digital forensics demonstrations and self-directed practical exercises. Some key techniques will be demonstrated during class, some others will be provided as independent study.

You will be expected to repeat and explore these activities in your own time.

You must install SIFT on your own device.

You will need to:
• Install VirtualBox (https://www.virtualbox.org)
• Download SIFT Workstation (https://www.sans.org/tools/sift-workstation/)
• Import the SIFT .ova file into VirtualBox

A full installation guide is provided below.

****NOTE for Apple Silicon users: There is no official SANS SIFT build for ARM.**
***Potential solution* -- Install Ubuntu 22.04 ARM64 and then gradually install the SIFT tools.**

- SIFT Installation Guide File

- The Linux command line for beginnersURL

- SANS SIFT Cheatsheet File

- **Administrative Matters**

- Announcements Forum
- Module Discussion Board Forum
- Introduction to the module File
- **Coursework**

Coursework instructions are provided here.

- **CW: Moodle Quiz**

The Moodle quiz will only be activated on 18th February 2026, 11:30-13:00. Only one hour is allotted to the quiz, so be sure to answer questions concisely. You are advised to review all questions prior to answering to ensure that you begin by answering the question(s) you feel most confident about, but do not dwell too long on any particular question.

The use of generative AI tools (e.g., Microsoft Copilot, ChatGPT, Grammarly) is not permitted for this assessment. More information regarding the use of generative AI in your studies can be found here.

- IY3609__1Quiz
- IY4609__1Quiz
- IY5609__1Quiz
- **IY5609O CW2: Term Paper**

This section and assessment is only relevant for students registered for the old 20-credit version of IY5609O.

- CW2 Guidance (V1) File

This document contains a few pointers to topic selection, formatting, and writing of term papers - this advice is relatively generic in nature, and should be equally applicable in many situations.

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CW2 Assignment Sheet (V1) File

You are advised to choose a topic for your term paper and start work on it as soon as possible; for this it is useful to review the syllabus to be informed about the scope of the module. Time management is an important aspect of study, and a well-chosen term paper topic presents an opportunity to study a topic in considerable depth, so deferring until late in the term is counterproductive.

▪

CW2 Submission PointAssignment

Not available unless: Your ID number is 101026492

- **Revision**

▪

CW1 - Discussion (V1) File

- **Wk 1 - Introduction to Digital Forensics**

Overview
On the first week of Digital Forensics the aim to introduce the structure of the module and what you should expect this term.

We will then discuss the main concepts of digital forensics, the different types of investigations and the importance of digital evidence.

Lecture Material

- Wk1 - Digital forensics concepts, investigations and digital evidence

Additional Resources
In this section you will find additional resources to support the topics covered each week.

- Crime Survey for England & Wales
- National Strategic Assessment 2025 of Serious and Organised Crime
- ACPO Guidelines
- Challenges to Digital Forensics
- Research Trends, Challenges, and Emerging Topics in Digital Forensics: A Review of Reviews

Play Video

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[Questions on Unit 1 Forum]

- **Wk 2 - Principles and Practices of Digital Evidence Handling**

Overview

This week the aim of the session is to cover relevant legislation and regulations to digital forensics, discuss the contact with the crime scene and expand on the collection and acquisition principles and considerations for different types of devices. Also, cover the importance the order of volatility plays in the collection of evidence.

Lecture Material

- Wk2 - Legislation and Regulations
- Wk2 - Investigation methodologies, collection & acquisition principles and considerations

Practical Material
In this section you will find the tutorial notes and/ or practical exercises relevant to this week's material, including any tasks for independent study after the end of the session.

- Disk Imaging Worksheet

Additional Resources
In this section you will find additional resources to support the topics covered each week.

- SoK- cross-border criminal investigations and digital evidence

- Interpol review of digital evidence 2016-2019
- A Review and Comparative Evaluation of Forensics Guidelines
- Quality standards for digital forensics: Learning from experience in England & Wales
- ISO/IEC 27037:2012(en) - Information technology — Security techniques — Guidelines for identification, collection, acquisition and preservation of digital evidence

- **Wk 3 - Data Storage Basics**

This week, we will explore the fundamental concepts underpinning forensic analysis and introduce the core principles of data storage and partitioning. We will then build on this by examining non-volatile storage concepts that are particularly important in digital forensics.

Lecture Material

- Wk3 - Data Storage Basics

Practical Material
In this section you will find tutorial notes and/ or practical exercises you will be working on every week, including any tasks for independent study.

- Data Storage Worksheet

Additional Resources
In this section you will find additional resources to support the topics covered each week.

- SSD Forensics - Where Do We Stand
- Solid State Drives - Data Reliability and Lifetime
- Forensic Images - For your Viewing Pleasure
- The Effect of TRIM Function on Data Recovery from SSD Solid-State Drives

- **Wk 4 - File Systems and File System Metadata**

This week, we will explore the fundamental concepts underpinning forensic analysis and introduce the main file systems, expand on FAT and NTFS and discuss the main areas on interest in digital forensics.

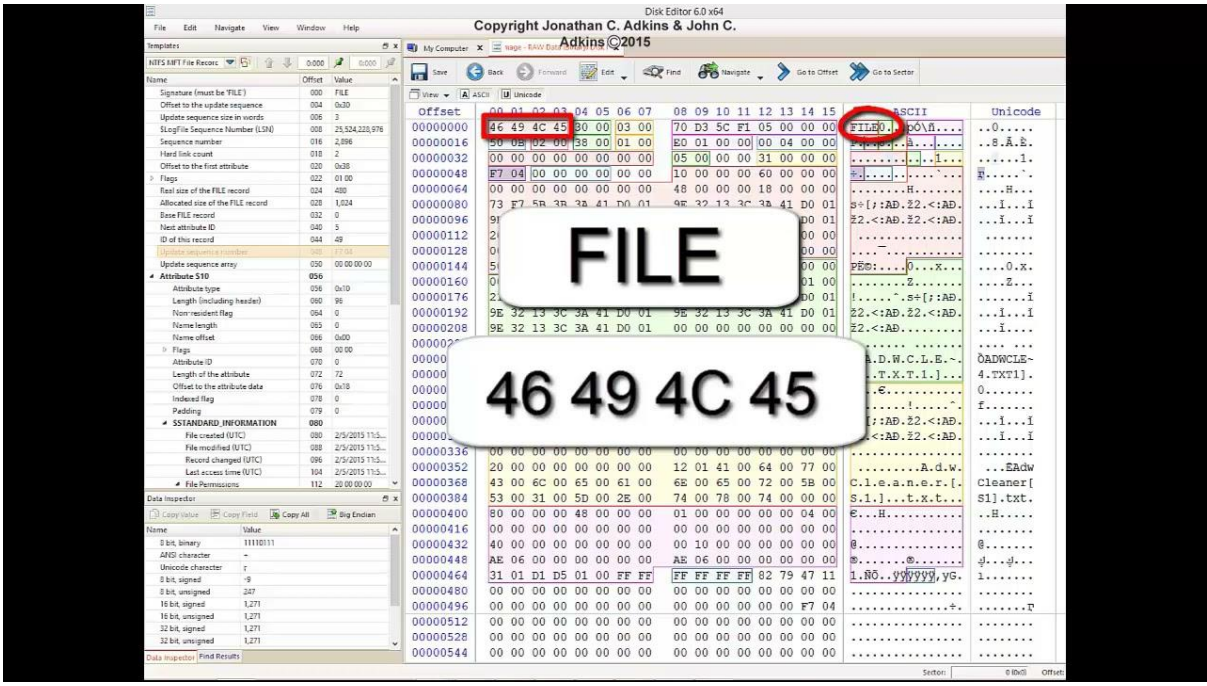
Lecture Material

- File Systems and File System Metadata

Practical Material
In this section you will find tutorial notes and/ or practical exercises you will be working on every week, including any tasks for independent study.

- Interpreting FAT Directory Entries Tutorial

Additional Resources
In this section you will find additional resources to support the topics covered each week.

- 

Play Video

- [Filesystems in the Linux Kernel]
- [NTFS Reference Sheet]
- [A contemporary investigation of NTFS file fragmentation]

- **Wk 5 - Host Forensics and Examination Approaches I**

This week, we will start exploring forensic analysis in more depth. We will cover the importance of file metadata, file signatures and expand on data carving.

Lecture Material

- [Host Forensics and Examination Approaches I]

Practical Material
In this section you will find tutorial notes and/ or practical exercises you will be working on every week, including any tasks for independent study.

- [Metadata and File carving exercises]

Additional Resources
In this section you will find additional resources to support the topics covered each week.

- [Data Carving Concepts]
- [Carving contiguous and fragmented files with fast object validation]
- [Beyond timestamps: Integrating implicit timing information into digital forensic timelines]

- **Wk 6 - Host Forensics and Examination Approaches II**

This week, we will continue exploring forensic analysis concepts. We will focus on the analysis and investigation concepts of Windows OS, the importance of the Windows registry and how analysing log files supports an investigation. We will also discuss the analysis of a timeline and the Windows encryption mechanisms.

Lecture Material

- **Host forensics and examination approaches II**

Practical Material
In this section you will find tutorial notes and/ or practical exercises you will be working on every week, including any tasks for independent study.

2. Autopsy Introduction : Please use *InCh05.img* for this exercise

3. Windows Registry Analysis : Please use *DR_HDD_image05.img* for this exercise

The forensic images required for this practical are available in the module's Teams channel. Please download them on your own device before beginning the exercise.

There are two tool options you may use for this practical:

- **Autopsy: You may use Autopsy either on Windows or within your SANS SIFT VM. Please refer here for further guidance on Autopsy.**
- **RegRipper: A registry analysis tool that is pre-installed in the SANS SIFT VM. Please refer to the provided guidance here for instructions on using RegRipper.**

Additional Resources
In this section you will find additional resources to support the topics covered each week.

- **Windows Incident Response Blog**
- **SANS DFIR Windows Artefacts Poster**
- **Windows registry harnesser for incident response and digital forensic analysis**
-

Unit 2 - Magnetic Storage Media, Volumes, and Encryption (V1) File

- **Wk 7 - Digital Forensics in Incident Response**

This week, we will discuss the incident response lifecycle and expand on DFIR and post incident forensics. We will also introduce live forensics and the principles of memory forensics.

Lecture Material

- **Digital Forensics in Incident Response - THIS IS AN UPDATED VERSION ON 06.03.2026**

Practical Material
In this section you will find tutorial notes and/ or practical exercises you will be working on every week, including any tasks for independent study.

The memory image required for this practical is available in the module's Teams channel. Please download it to your own device before beginning the exercise.

- Memory Analysis

Additional Resources
In this section you will find additional resources to support the topics covered each week.

- Reinforcement learning for an efficient and effective malware investigation during cyber incident response
- How to Use Volatility for Memory Forensics and Analysis
- Memory Samples **for practise**
- SANS Memory Forensics Cheat Sheet
-

Unit 2 - FAT File Systems (V1)

- **Wk 8 - Network Forensics Principles and Practice**

This week, we will explore review IPv4 and IPv6 information from a forensic perspective and discuss where network based evidence can be identified, how the analysis is approached and the types of network based evidence.

Lecture Material

- Network Forensics Principles and Practice

Practical Material
In this section you will find tutorial notes and/ or practical exercises you will be working on every week, including any tasks for independent study.

The IIS log file and the network capture required for these practicals are available in the module's Teams channel. Please download them to your own device before beginning the exercises.

- Log Analysis
- Network Forensics - Wireshark

Additional Resources
In this section you will find additional resources to support the topics covered each week.

- SANS Network Forensics Poster
- Application of artificial intelligence to network forensics: Survey, challenges and future directions
-

Unit 2 - Reconstructing FAT Structures (V1) File

- **Wk 9 - Malware Analysis for Digital Forensics**

This week, we will explore the different types of malware and how we approach the analysis and investigation of malicious software.

Lecture Material

- Malware Analysis for Digital Forensics

Practical Material
In this section you will find tutorial notes and/ or practical exercises you will be working on every week, including any tasks for independent study.

- Malware Analysis

Additional Resources
In this section you will find additional resources to support the topics covered each week.

- SANS Hunt Evil Poster
- NCSC Malware Analysis Reports
- Technical Approaches to Uncovering and Remediating Malicious Activity
-

Unit 3 - The Microsoft Windows Storage Architecture (V1) File

- **Wk 10 - Mobile Forensics Principles**

This week, we will explore mobile forensics and discuss the unique challenges they bring in a digital forensic investigation.

Lecture Material

- Mobile Forensics Principles

Practical Material
In this section you will find tutorial notes and/ or practical exercises you will be working on every week, including any tasks for independent study.

- **This week your task is to respond to the following research questions. Please keep structured notes of your analysis and ensure that your responses are informed by relevant academic and scientific literature.**

1. To what extent can mobile device artefacts reliably establish user activity and intent? You may want to use specific application examples.

2. How does the shift from device-centric storage to cloud-centric architectures affect mobile forensic investigations?

3. How have modern mobile operating system security mechanisms changed forensic acquisition capabilities?

4. How reliable are mobile application artefacts as evidence of user behaviour?

Additional Resources
In this section you will find additional resources to support the topics covered each week.

- A study of the data remaining on second hand mobile devices in the UK
- Forensic analysis of Telegram Messenger on iOS smartphones
- Study of identifying and managing the potential evidence for effective Android forensics
-

Unit 3 - The Windows NTFS File System (V1)

- **Wk 11 - Forensic Process, Documentation and Reporting**

This week, we will revisit the forensic investigation process, discuss the different types of tools available and focus on the reporting of forensic evidence.

Lecture Material

- Forensic Process, Documentation and Reporting

Practical Material
In this section you will find tutorial notes and/ or practical exercises you will be working on every week, including any tasks for independent study.

- Craft an Investigation Report

Additional Resources
In this section you will find additional resources to support the topics covered each week.

- Example of an expert witness digital forensics report
- Who is the digital forensic expert and what is their expertise?
- Certification, registration and assessment of digital forensic experts: The UK experience
-

Unit 3 - File and Volume Encryption (V1)

- **Block Mode Week Tasks - Block Mode students ONLY**
  - **Day 2 - Data Storage Basics**

Please complete these tasks before the session:

1. Watch the lecture recording from Wk3 - Data Storage Basics, available on Panopto.

2. Read the paper titled "The Effect of TRIM Function on Data Recovery from SSD Solid-State Drives", which is available under the 'Wk3 - Data Storage Basics' materials on Moodle.

3. Submit a 200-word response to the question provided in the attached Moodle Quiz. Your response should demonstrate engagement with both the lecture content and the assigned reading.

- Data Storage BasicsQuestionnaire

  - Day 5 - Network Forensics Principles and Practice

Please complete these tasks before the session:

1. Study the Wk8 - Network Forensics Principles and Practice lecture notes.

2. Watch the Network Forensics: Uncover Cyber Threats Through Network Analysis video

Play Video

3. Submit short responses to the questions provided in the attached Moodle Quiz. Your responses should demonstrate engagement with the lecture content.

- Network forensicsQuestionnaire

- Breaking and Fixing Apple AirDrop File